

Moving Charges and Magnetism

Biot Savart Law, Magnetic field due to current carrying wire

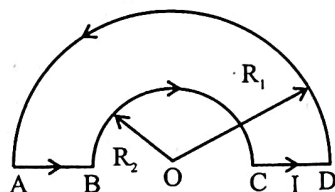
1. The percentage increase in magnetic field (B) when space within a current carrying solenoid is filled with magnesium (magnetic susceptibility $\chi_{mg} = 1.2 \times 10^{-5}$) is : [07 April, 2025 (Shift-I)]

- (a) $\frac{6}{5} \times 10^{-3}\%$ (b) $\frac{5}{6} \times 10^{-5}\%$
(c) $\frac{5}{6} \times 10^{-4}\%$ (d) $\frac{5}{3} \times 10^{-5}\%$

2. A loop $ABCD$, carrying current $I = 12\text{ A}$, is placed in a plane, consists of two semi-circular segments of radius $R_1 = 6\pi\text{ m}$ and $R_2 = 4\pi\text{ m}$. The magnitude of the resultant magnetic field at center O is $k \times 10^{-7}\text{ T}$. The value of k is _____.

(Given $\mu_0 = 4\pi \times 10^{-7}\text{ TmA}^{-1}$)

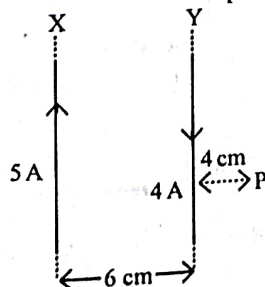
[03 April, 2025 (Shift-I)]



3. Two long parallel wires X and Y , separated by a distance of 6 cm , carry currents of 5 A and 4 A , respectively, in opposite directions as shown in the figure. Magnitude of the resultant magnetic field at point P at a distance of 4 cm from wire Y is $x \times 10^{-5}\text{ T}$. The value of x is _____.

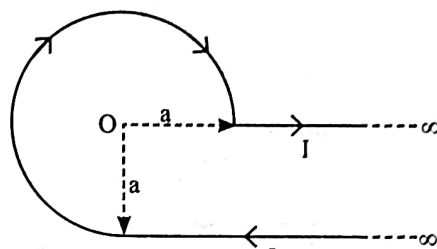
Take permeability of free space as $\mu_0 = 4\pi \times 10^{-7}\text{ SI units}$.

[22 Jan, 2025 (Shift-II)]



4. A current of 5 A exists in a square loop of side $\frac{1}{\sqrt{2}}\text{ m}$. Then the magnitude of the magnetic field B at the centre of the square loop will be $p \times 10^{-6}\text{ T}$, where, value of p is _____. [24 Jan, 2025 (Shift-I)]
[Take $\mu_0 = 4\pi \times 10^{-7}\text{ TmA}^{-1}$].

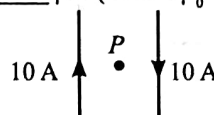
5.



An infinite wire has a circular bend of radius a , and carrying a current I as shown in figure. The magnitude of magnetic field at the origin O of the arc is given by: [28 Jan, 2025 (Shift-II)]

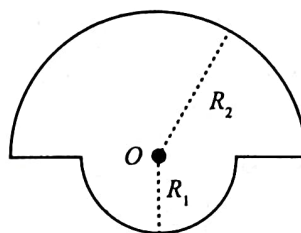
- (a) $\frac{\mu_0 I}{4\pi a} \left[\frac{3\pi}{2} + 2 \right]$ (b) $\frac{\mu_0 I}{4\pi a} \left[\frac{3\pi}{2} + 1 \right]$
(c) $\frac{\mu_0 I}{2\pi a} \left[\frac{\pi}{2} + 2 \right]$ (d) $\frac{\mu_0 I}{4\pi a} \left[\frac{\pi}{2} + 1 \right]$

6. Two long, straight wires carry equal currents in opposite directions as shown in figure. The separation between the wires is 5.0 cm . The magnitude of the magnetic field at a point P midway between the wires is _____ $\mu\text{ T}$. (Given: $\mu_0 = 4\pi \times 10^{-7}\text{ TmA}^{-1}$)



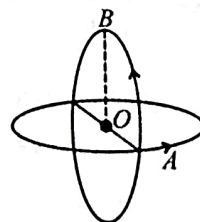
[27 Jan, 2024 (Shift-I)]

7. The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2\pi\text{ m}$ and $R_2 = 4\pi\text{ m}$ carrying current $I = 4\text{ A}$ as per figure given below is $\alpha \times 10^{-7}\text{ T}$. The value of α is _____. (Centre O is common for all segments)



[27 Jan, 2024 (Shift-II)]

8. Two insulated circular loop A and B radius ' a ' carrying a current of ' I ' in the anti clockwise direction as shown in figure. The magnitude of the magnetic induction at the centre will be :



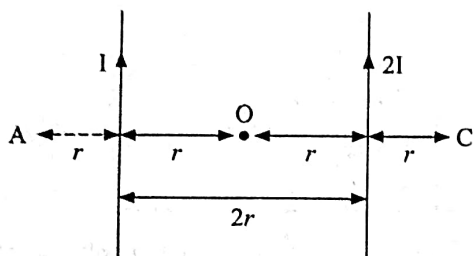
[30 Jan, 2024 (Shift-I)]

- (a) $\frac{\sqrt{2}\mu_0 I}{a}$ (b) $\frac{\mu_0 I}{2a}$ (c) $\frac{\mu_0 I}{\sqrt{2}a}$ (d) $\frac{2\mu_0 I}{a}$

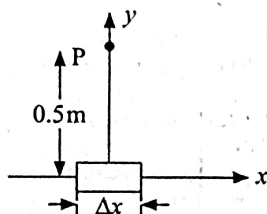
9. The current of 5A flows in a square loop of sides 1 m is placed in air. The magnetic field at the centre of the loop is $X\sqrt{2} \times 10^{-7}$ T. The value of X is _____. [30 Jan, 2024 (Shift-II)]

10. A regular polygon of 6 sides is formed by bending a wire of length 4π meter. If an electric current of $4\pi\sqrt{3}$ A is flowing through the sides of the polygon, the magnetic field at the centre of the polygon would be $x \times 10^{-3}$ T. The value of x is _____. [1 Feb, 2024 (Shift-I)]

11. Two parallel long current carrying wire separated by a distance $2r$ are shown in the figure. The ratio of magnetic field at A to the magnetic field produced at C is $\frac{x}{7}$. The value of x is _____. [04 April, 2024 (Shift-II)]

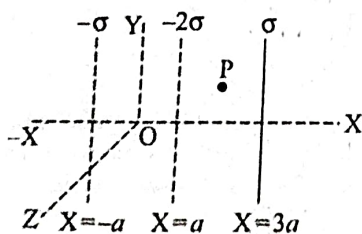


12. An element $\Delta l = \Delta x \hat{i}$ is placed at the origin and carries a large current $I = 10$ A. The magnetic field on the y -axis at a distance of 0.5 m from the elements Δx of 1 cm length is: [06 April, 2024 (Shift-I)]

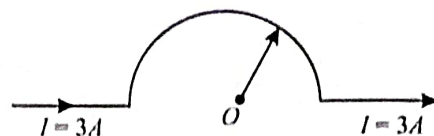


- (a) 4×10^{-8} T (b) 8×10^{-8} T
(c) 12×10^{-8} T (d) 10×10^{-8} T

13. Three infinitely long charged thin sheets are placed as shown in figure. The magnitude of electric field at the point P is $\frac{x\sigma}{\epsilon_0}$. The value of x is _____. [06 April, 2024 (Shift-I)]
(all quantities are measured in SI units).

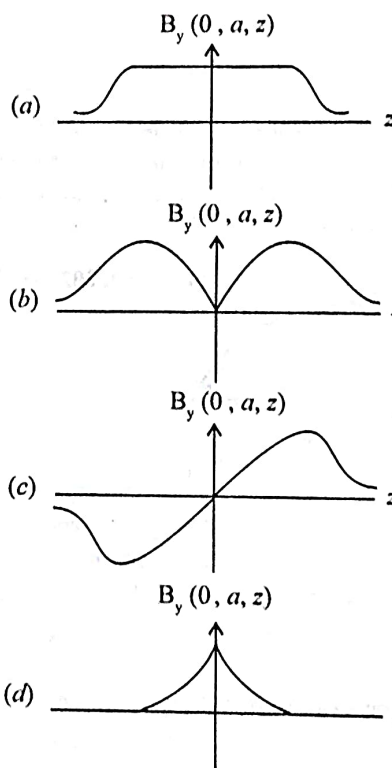
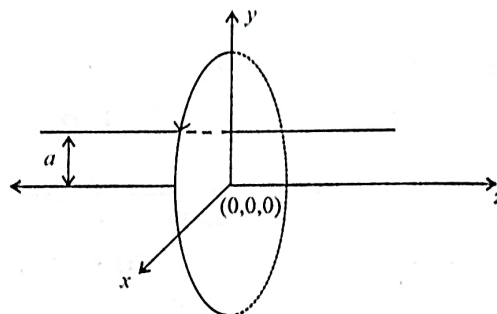


14. As shown in the figure, a long straight conductor with semicircular arc of radius $\frac{\pi}{10}$ m is carrying current $I = 3$ A. The magnitude of the magnetic field at the center O of the arc is: (The permeability of the vacuum $= 4\pi \times 10^{-7}$ N A⁻²) [1 Feb, 2023 (Shift-II)]

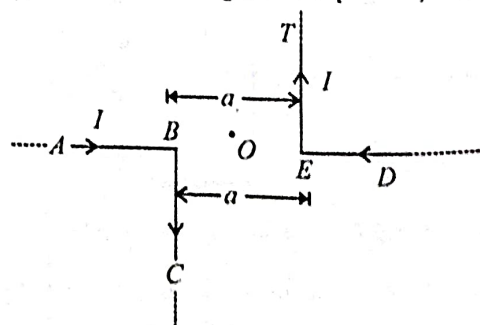


- (a) 6μ T (b) 1μ T (c) 4μ T (d) 3μ T

15. A single current carrying loop of wire carrying current I flowing in anticlockwise direction as seen from +ve z direction and lying in xy plane is shown in figure. The plot of $\hat{j}$ component of magnetic field (B_y) at a distance ' a ' (less than radius of the coil) and on yz plane vs z coordinate look like [29 Jan, 2023 (Shift-I)]

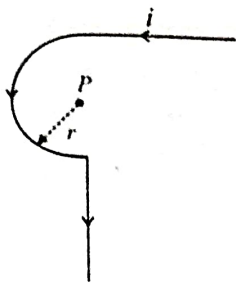


16. The magnitude of magnetic induction at mid-point O due to current arrangement as shown in the Fig will be: [29 Jan, 2023 (Shift-I)]



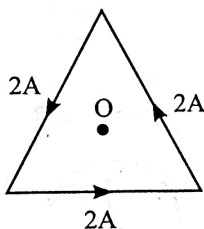
- (a) $\frac{\mu_0 I}{2\pi a}$ (b) 0 (c) $\frac{\mu_0 I}{4\pi a}$ (d) $\frac{\mu_0 I}{\pi a}$

17. Find the magnetic field at the point P in figure. The curved portion is a semicircle connected to two long straight wires. [1 Feb, 2023 (Shift-I)]



- (a) $\frac{\mu_0 i}{2r} \left(1 + \frac{2}{\pi}\right)$ (b) $\frac{\mu_0 i}{2r} \left(1 + \frac{1}{\pi}\right)$
 (c) $\frac{\mu_0 i}{2r} \left(\frac{1}{2} + \frac{1}{2\pi}\right)$ (d) $\frac{\mu_0 i}{2r} \left(\frac{1}{2} + \frac{1}{\pi}\right)$

18. As shown in the figure, a current of 2A flows in an equilateral triangle of side $4\sqrt{3}$ cm. The magnetic field at the centroid O of the triangle is: [30 Jan, 2023 (Shift-II)]



(Neglect the effect of earth's magnetic field.)

- (a) $4\sqrt{3} \times 10^{-4}$ T (b) $4\sqrt{3} \times 10^{-5}$ T
 (c) $\sqrt{3} \times 10^{-4}$ T (d) $3\sqrt{3} \times 10^{-5}$ T

19. A circular loop of radius r is carrying current I A. The ratio of magnetic field at the centre of circular loop and at a distance r from the center of the loop on its axis is: [24 Jan, 2023 (Shift-I)]

- (a) $1:3\sqrt{2}$ (b) $3\sqrt{2}:2$ (c) $2\sqrt{2}:1$ (d) $1:\sqrt{2}$

20. Match List-I with List-II [25 Jan, 2023 (Shift-I)]

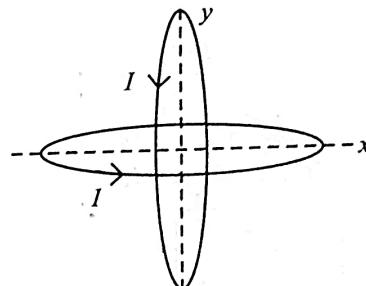
List - I (Current configuration)	List - II (Magnetic field at point O)
A 	I. $B_0 = \frac{\mu_0 I}{4\pi r} [\pi + 2]$
B 	II. $B_0 = \frac{\mu_0 I}{4r}$
C 	III. $B_0 = \frac{\mu_0 I}{2\pi r} [\pi - 1]$
D 	IV. $B_0 = \frac{\mu_0 I}{4\pi r} [\pi + 1]$

Choose the correct answer from the option given below:

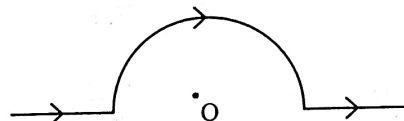
- (a) A-III, B-IV, C-I, D-II (b) A-I, B-III, C-IV, D-II
 (c) A-III, B-I, C-IV, D-II (d) A-II, B-I, C-IV, D-III

21. Two long parallel wires carrying currents 8 A and 15 A in opposite directions are placed at a distance of 7 cm from each other. A point P is equidistant from both the wires such that the lines joining the point P to the wires are perpendicular to each other. Magnitude of the magnetic field at P is $\times 10^{-4}$ T. (Given: $\sqrt{2} = 1.4$) [25 Jan, 2023 (Shift-II)]

22. Two identical circular wires of radius 20 cm and carrying current $\sqrt{2}$ A are placed in perpendicular planes as shown in figure. The net magnetic field at the centre of the circular wire is $\times 10^{-8}$ T. (Take $\pi = 3.14$) [6 April, 2023 (Shift-I)]



23. A straight wire carrying a current of 14 A is bent into a semicircular arc of radius 2.2 cm as shown in the figure. The magnetic field produced by the current at the centre (O) of the arc is $\times 10^{-4}$ T [10 April, 2023 (Shift-II)]



24. An electron in a hydrogen atom revolves around its nucleus with a speed of 6.76×10^6 ms $^{-1}$ in an orbit of radius 0.52 Å. The magnetic field produced at the nucleus of the hydrogen atom is $\times 10^{-4}$ T. [15 April, 2023 (Shift-I)]

25. The ratio of magnetic field at the centre of a current carrying coil of radius r to the magnetic field at distance r from the centre of coil on its axis is $\sqrt{x}:1$. The value of x is $\times 10^{-4}$ T. [8 April, 2023 (Shift-II)]

26. Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is 300×10^{-4} T. The equal current flowing in the two conductors is: [29 June, 2022 (Shift-II)]

- (a) 30A in the same direction.
 (b) 30 A in the opposite direction.
 (c) 60A in the opposite direction.
 (d) 300A in the opposite direction.

27. The magnetic field at the centre of current carrying circular loop is B_1 . The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2 . The value of B_1/B_2 will be [28 July, 2022 (Shift-II)]

- (a) 9:4 (b) $12:\sqrt{5}$ (c) 8:1 (d) $5:\sqrt{3}$

28. The magnetic field at the centre of a circular coil of radius r , due to current I flowing through it, is B . The magnetic field at a point along the axis at a distance $r/2$ from the centre is: [24 June, 2022 (Shift-I)]

- (a) $B/2$ (b) $2B$ (c) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (d) $\left(\frac{1}{\sqrt{3}}\right)^3 B$

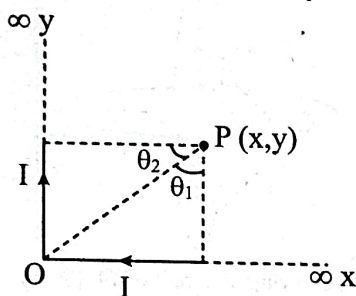
29. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii ' a ' and ' b ' respectively. Find the magnetic field at centre, when a current I passes through coil:

[31 Aug, 2021 (Shift-I)]

- (a) $\frac{\mu_0 I}{8} \left[\frac{a+b}{a-b} \right]$ (b) $\frac{\mu_0 I N}{2(b-a)} \log_e \left(\frac{b}{a} \right)$
 (c) $\frac{\mu_0 I}{4(a-b)} \left[\frac{1}{a} - \frac{1}{b} \right]$ (d) $\frac{\mu_0 I}{8} \left(\frac{a-b}{a+b} \right)$

30. There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductor is 1 : 1. The magnetic field at point P is

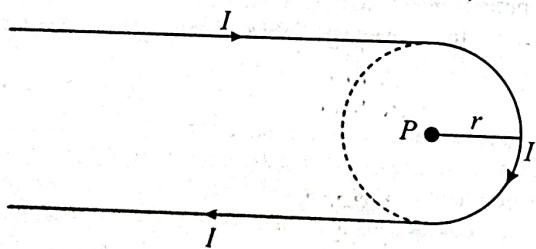
[1 Sept, 2021 (Shift-II)]



- (a) $\frac{\mu_0 I}{4\pi xy} [\sqrt{x^2 + y^2} - (x+y)]$ (b) $\frac{\mu_0 I}{4\pi xy} [\sqrt{x^2 + y^2} + (x+y)]$
 (c) $\frac{\mu_0 I xy}{4\pi} [\sqrt{x^2 + y^2} - (x+y)]$ (d) $\frac{\mu_0 I xy}{4\pi} [\sqrt{x^2 + y^2} + (x+y)]$

31. A hairpin like shape as shown in figure is made by bending a long current carrying wire. What is the magnitude of a magnetic field at point P which lies on the centre of the semicircle?

[17 March, 2021 (Shift-II)]



- (a) $\frac{\mu_0 I}{2\pi r} (2 - \pi)$ (b) $\frac{\mu_0 I}{4\pi r} (2 - \pi)$
 (c) $\frac{\mu_0 I}{2\pi r} (2 + \pi)$ (d) $\frac{\mu_0 I}{4\pi r} (2 + \pi)$

32. A current of 1.5 A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is:

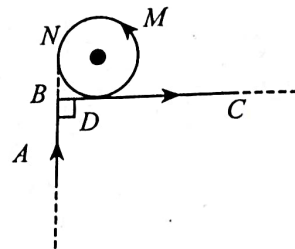
(Assume that the current is flowing in the clockwise direction.)

[31 Aug, 2021 (Shift-II)]

- (a) $2\sqrt{3} \times 10^{-3} T$, outside the plane of triangle
 (b) $2\sqrt{3} \times 10^{-3} T$, inside the plane of triangle
 (c) $3 \times 10^{-3} T$, inside the plane of triangle
 (d) $3 \times 10^{-3} T$, outside the plane of triangle

33. A very long wire ABDMNDC is shown in figure carrying current I . AB and BC parts are straight, long and at right angle. At D wire forms a circular turn $DMND$ of radius R . AB , BC parts are tangential to circular turn at N and D . Magnetic field at the centre of circle is:

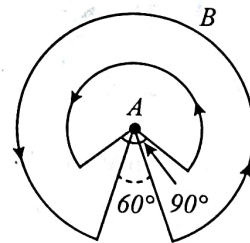
[8 Jan, 2020 (Shift-II)]



- (a) $\frac{\mu_0 I}{2R}$ (b) $\frac{\mu_0 I}{2\pi R} \left(\pi - \frac{1}{\sqrt{2}} \right)$
 (c) $\frac{\mu_0 I}{2\pi R} (\pi + 1)$ (d) $\frac{\mu_0 I}{2\pi R} \left(\pi + \frac{1}{2} \right)$

34. A wire A , bent in the shape of an arc of a circle, carrying a current of 2 A and having radius 2 cm and another wire B , also bent in the shape of arc of a circle, carrying a current of 3 A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common centre O is

[04 Sept, 2020 (Shift-I)]



- (a) 2 : 5 (b) 6 : 5 (c) 4 : 6 (d) 6 : 4

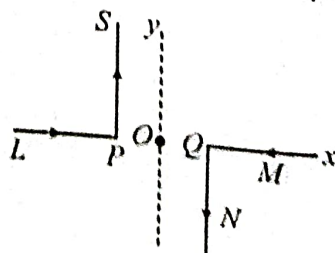
35. One of the two identical conducting wires of length L is bent in the form of a circular loop and the other one into a circular coil of N identical turns. If the same current is passed in both, the ratio of the magnetic field at the central of the loop (B_L) to that at the centre of the coil (B_C), i.e. $\frac{B_L}{B_C}$ will be

[9 Jan, 2019 (Shift-II)]

- (a) N (b) $\frac{1}{N}$
 (c) N^2 (d) $\frac{1}{N^2}$

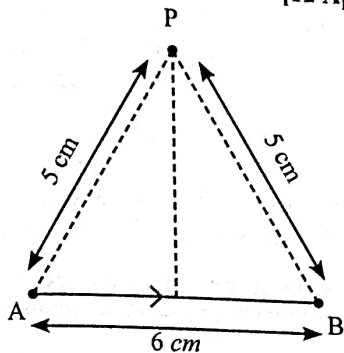
36. As shown in the figure, two infinitely long, identical wires are bent by 90° and placed in such a way that the segments LP and QM are along the x -axis, while segments PS and QN are parallel to the y -axis. If $OP = OQ = 4$ cm, and the magnitude of the magnetic field at O is $10^{-4} T$, and the two wires carry equal current (see figure), the magnitude of the current in each wire and the direction of the magnetic field at O will be ($\epsilon_0 = 4\pi \times 10^{-7} NA^{-2}$)

[12 Jan, 2019 (Shift-I)]



- (a) 20 A, perpendicular out of the page.
 (b) 40 A, perpendicular out of the page.
 (c) 20 A, perpendicular into the page.
 (d) 40 A, perpendicular into the page.

37. Find the magnetic field at point P due to a straight line segment AB of length 6 cm carrying a current of 5 A. (See figure) ($\infty = 4\pi \times 10^{-7} \text{ N/A}^2$)
 [12 April, 2019 (Shift-II)]

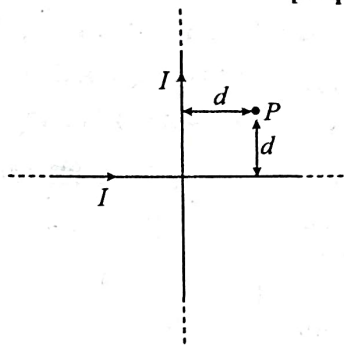


- (a) $35.0 \times 10^{-5} \text{ T}$ (b) $3.0 \times 10^{-5} \text{ T}$
 (c) $2.5 \times 10^{-5} \text{ T}$ (d) $1.5 \times 10^{-5} \text{ T}$

38. A thin ring of 10 cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of $40\pi \text{ rads}^{-1}$ about its axis, perpendicular to its plane. If the magnetic field at its centre is $3.8 \times 10^{-9} \text{ T}$, then the charge carried by the ring is close to ($\infty = 4\pi \times 10^{-7} \text{ N/A}^2$).
 [12 April, 2019 (Shift-I)]

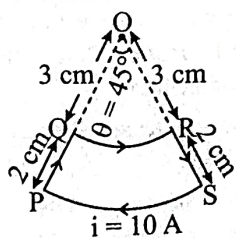
- (a) $2 \times 10^{-6} \text{ C}$ (b) $7 \times 10^{-6} \text{ C}$ (c) $4 \times 10^{-5} \text{ C}$ (d) $3 \times 10^{-7} \text{ C}$

39. Two very long, straight and insulated wires are kept at 90° angle from each other in xy -plane as shown in the figure. These wires carry currents of equal magnitude I , whose directions are shown in the figure. The net magnetic field at point P will be
 [8 April, 2019 (Shift-II)]



- (a) Zero (b) $\frac{\mu_0 I}{\pi d} (\hat{z})$
 (c) $-\frac{\mu_0 I}{2\pi d} (\hat{x} + \hat{y})$ (d) $\frac{\mu_0 I}{2\pi d} (\hat{x} + \hat{y})$

40. A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to:
 [9 Jan, 2019 (Shift-I)]



- (a) $39.0 \times 10^{-7} \text{ T}$ (b) $1.5 \times 10^{-7} \text{ T}$
 (c) $1.5 \times 10^{-5} \text{ T}$ (d) $1.0 \times 10^{-5} \text{ T}$

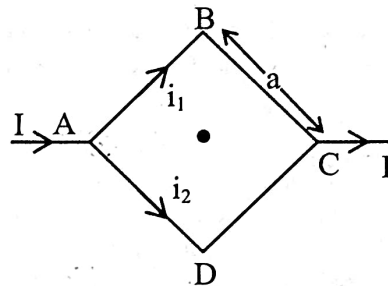
Ampere's Circuital Law and Its Applications

41. Let B_1 be the magnitude of magnetic field at center of a circular coil of radius R carrying current I . Let B_2 be the magnitude of magnetic field at an axial distance ' x ' from the center. For $x : R = 3 : 4$, $\frac{B_2}{B_1}$ is:

[02 April, 2025 (Shift-I)]

- (a) 4:5 (b) 16:25
 (c) 64:125 (d) 25:16

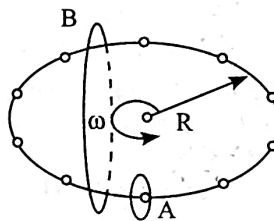
42. Figure shows a current carrying square loop ABCD of edge length is ' a ' lying in a plane. If the resistance of the ABC part is r and that of ADC part is $2r$, then the magnitude of the resultant magnetic field at centre of the square loop is



[08 April, 2025 (Shift-II)]

- (a) $\frac{3\pi\mu_0 I}{\sqrt{2}a}$ (b) $\frac{\mu_0 I}{2\pi a}$
 (c) $\frac{\sqrt{2}\mu_0 I}{3\pi a}$ (d) $\frac{2\mu_0 I}{3\pi a}$

43.

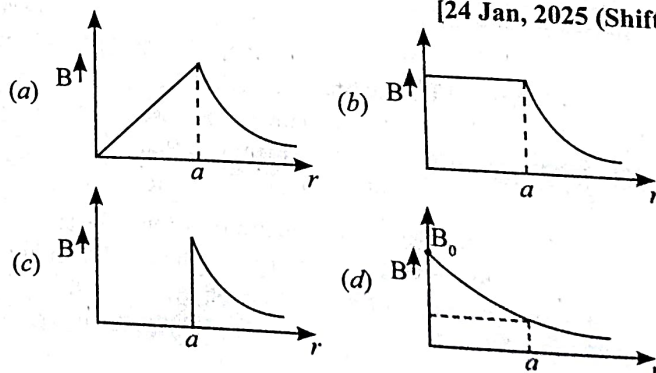


N equally spaced charges each of value q , are placed on a circle of radius R . The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop B encloses the whole circle where as a smaller Amperian loop A encloses a small segment. The difference between enclosed currents, $I_A - I_B$, for the given Amperian loops is [24 Jan, 2025 (Shift-II)]

- (a) $\frac{N}{\pi} q \omega$ (b) $\frac{2\pi}{N} q \omega$ (c) $\frac{N}{2\pi} q \omega$ (d) $\frac{N^2}{2\pi} q \omega$

44. A long straight wire of a circular cross-section with radius ' a ' carries a steady current I . The current I is uniformly distributed across this cross-section. The plot of magnitude of magnetic field B with distance r from the centre of the wire is given by

[24 Jan, 2025 (Shift-II)]



A tightly wound long solenoid carries a current of 1.5 A. An electron is executing uniform circular motion inside the solenoid with a time period of 75 ns. The number of turns per metre in the solenoid is _____. [Take mass of electron $m_e = 9 \times 10^{-31}$ kg, charge of electron $|q_e| = 1.6 \times 10^{-19}$ C, $\mu_0 = 4\pi \times 10^{-7} \frac{\text{N}}{\text{A}^2}$, $1 \text{ ns} = 10^{-9} \text{ s}$]

[24 Jan, 2025 (Shift-II)]

Consider a long straight wire of a circular cross-section (radius a) carrying a steady current I . The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, any where due to the wire, will be

[29 Jan, 2025 (Shift-I)]

(a) $[a/2, 3a]$ (b) $[a/4, 2a]$ (c) $[a/4, 3a/2]$ (d) $[a/2, 2a]$

7. The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} Tesla. If the solenoid carries a current of 0.29 A, then the length of the solenoid is _____ π cm.

[29 Jan, 2025 (Shift-II)]

8. Two circular coils P and Q of 100 turns each have same radius of π cm. The currents in P and R are 1 A and 2 A respectively. P and Q are placed with their planes mutually perpendicular with their centers coincide. The resultant magnetic field induction at the center of the coils is $\sqrt{x}mT$, where $x =$ _____.

[Use $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$]

[31 Jan, 2024 (Shift-II)]

49. In a co-axial straight cable, the central conductor and the outer conductor carry equal currents in opposite directions. The magnetic field is zero.

[05 April, 2024 (Shift-I)]

(a) inside the outer conductor (b) in between the two conductors (c) outside the cable (d) inside the inner conductor

50. A solenoid of length 0.5 m has a radius of 1 cm and is made up of ' m ' number of turns. It carries a current of 5A. If the magnitude of the magnetic field inside the solenoid is $6.28 \times 10^{-3} \text{ T}$, then the value of m is :

[05 April, 2024 (Shift-II)]

51. A long straight wire of radius a carries a steady current I . The current is uniformly distributed across its cross section. The ratio of the magnetic field at $a/2$ and $2a$ from axis of the wire is:

[08 April, 2024 (Shift-II)]

(a) 1 : 4 (b) 4 : 1 (c) 1 : 1 (d) 3 : 4

52. Given below are two statements:

Statement (I): When currents vary with time, Newton's third law is valid only if momentum carried by the electromagnetic field is taken into account.

Statement (II): Ampere's circuital law does not depend on Biot-Savart's law.

In the light of the above statements, choose the correct answer from the options given below:

[09 April, 2024 (Shift-I)]

(a) Both Statement I and Statement II are false.

(b) Statement I is true but Statement II is false.

(c) Statement I is false but Statement II is true.

(d) Both Statement I and Statement II are true.

53. The magnetic intensity at the centre of a long current carrying solenoid is found to be $1.6 \times 10^3 \text{ Am}^{-1}$. If the number of turns is 8 per cm, then the current flowing through the solenoid is _____ A.

[8 April, 2023 (Shift-I)]

54. A charge particle moves along circular path in a uniform magnetic field in a cyclotron. The kinetic energy of the charge particle increases to 4 times its initial value. What will be the ratio of new radius to the original radius of circular path of the charge particle:

[29 June, 2022 (Shift-I)]

(a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 1 : 4

55. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

[25 June, 2022 (Shift-II)]

(a) B (b) $2B$ (c) $4B$ (d) $\frac{B}{2}$

56. A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and $2a$, respectively from the axis of the wire is

[9 Jan, 2020 (Shift-I)]

(a) $\frac{1}{2}$ (b) 2 (c) $\frac{3}{2}$ (d) $\frac{2}{3}$

57. A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5 T. If an electric field of 100 V/m makes it to move in a straight path, then the mass of the particle is (given charge of electron = $1.6 \times 10^{-19} \text{ C}$)

[9 Jan, 2019 (Shift-II)]

(a) $48.1 \times 10^{-31} \text{ kg}$ (b) $1.6 \times 10^{-27} \text{ kg}$
(c) $49.6 \times 10^{-19} \text{ kg}$ (d) $2.0 \times 10^{-24} \text{ kg}$

Motion of Charged Particle in Magnetic Field, Magnetic Force

58. An electron with mass ' m ' with an initial velocity ($t = 0$) $\vec{v} = v_0 \hat{i}$ ($v_0 > 0$) enters a magnetic field $\vec{B} = B_0 \hat{j}$. If the initial de-Broglie wavelength at $t = 0$ is λ_0 then its value after time ' t ' would be:

[02 April, 2025 (Shift-II)]

(a) $\frac{\lambda_0}{\sqrt{1 - \frac{e^2 B_0^2 t^2}{m^2}}}$ (b) $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 B_0^2 t^2}{m^2}}}$
(c) $\lambda_0 \sqrt{1 + \frac{e^2 B_0^2 t^2}{m^2}}$ (d) λ_0

59. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If Oxygen ion (O^{2-}) and Hydrogen ion (H^+) enter normal to the magnetic field with equal momentum, then the path of O^{2-} ion has a smaller curvature than that of H^+ .

Reason R: A proton with same linear momentum as an electron will form a path of smaller radius of curvature on entering a uniform magnetic field perpendicularly.

In the light of the above statements, choose the correct answer from the options given below

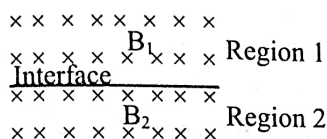
[03 April, 2025 (Shift-II)]

(a) A is true but R is false
(b) Both A and R are true but R is NOT the correct explanation of A
(c) A is false but R is true
(d) Both A and R are true and R is the correct explanation of A

60. A particle of charge $1.6\mu\text{C}$ and mass $16\mu\text{g}$ is present in a strong magnetic field of 6.28 T . The particle is then fired perpendicular to magnetic field. The time required for the particle to return to original location for the first time is _____ s. ($\pi = 3.14$)

[04 April, 2025 (Shift-II)]

61. Uniform magnetic fields of different strengths (B_1 and B_2), both normal to the plane of the paper exist as shown in the figure. A charged particle of mass m and charge q , at the interface at an instant, moves into the region 2 with velocity v and returns to the interface. It continues to move into region 1 and finally reaches the interface. What is the displacement of the particle during this movement along the interface?

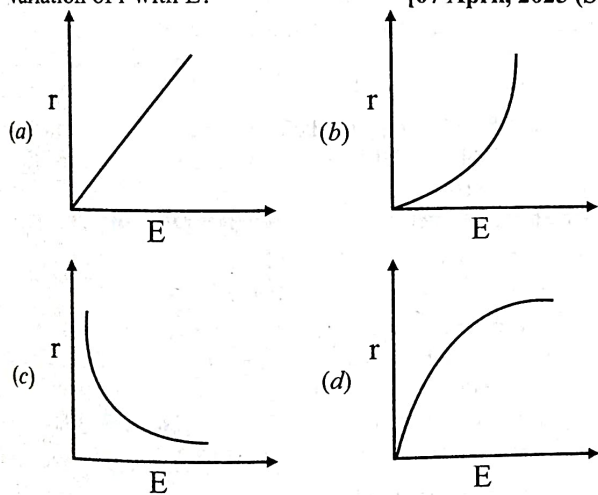


(Consider the velocity of the particle to be normal to the magnetic field and $B_2 > B_1$)

[07 April, 2025 (Shift-I)]

- (1) $\frac{mv}{qB_1} \left(1 - \frac{B_2}{B_1}\right) \times 2$ (2) $\frac{mv}{qB_1} \left(1 - \frac{B_1}{B_2}\right)$
 (3) $\frac{mv}{qB_1} \left(1 - \frac{B_2}{B_1}\right)$ (4) $\frac{mv}{qB_1} \left(1 - \frac{B_1}{B_2}\right) \times 2$
62. A particle of charge q , mass m and kinetic energy E enters in magnetic field perpendicular to its velocity and undergoes a circular arc of radius(r). Which of the following curves represents the variation of r with E ?

[07 April, 2025 (Shift-I)]



63. A proton is moving undeflected in a region of crossed electric and magnetic fields at a constant speed of $2 \times 10^5\text{ ms}^{-1}$. When the electric field is switched off, the proton moves along a circular path of radius 2 cm . The magnitude of electric field is $x \times 10^4\text{ N/C}$. The value of x is _____.

Take the mass of the proton = $1.6 \times 10^{-27}\text{ kg}$.

[22 Jan, 2025 (Shift-II)]

64. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): A electron in a certain region of uniform magnetic field is moving with constant velocity in a straight line path.

Reason (R): The magnetic field in that region is along the direction of velocity of the electron. In the light of the above statements, choose the correct answer from the options given below :

[24 Jan, 2025 (Shift-II)]

- (a) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
 (b) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (c) (A) is true but (R) is false
 (d) (A) is false but (R) is true

65. Consider a long thin conducting wire carrying a uniform current I . A particle having mass " M " and charge " q " is released at a distance " a " from the wire with a speed v_0 along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is [μ_0 is vacuum permeability]

[28 Jan, 2025 (Shift-I)]

- (a) $a \left[1 - \frac{mv_0}{q\mu_0 I}\right]$ (b) $\frac{a}{2}$
 (c) $a \left[1 - \frac{mv_0}{2q\mu_0 I}\right]$ (d) $ae^{\frac{-4\pi mv_0}{q\mu_0 I}}$

66. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, then the region of space may have:

[27 Jan, 2024 (Shift-I)]

- (A) $E = 0, B = 0$ (B) $E = 0, B \neq 0$
 (C) $E \neq 0, B = 0$ (D) $E \neq 0, B \neq 0$

Choose the most appropriate answer from the options given below:

- (a) (A), (B) and (C) only (b) (A), (C) and (D) only
 (c) (A), (B) and (D) only (d) (B), (C) and (D) only

67. Two particles X and Y having equal charges are being accelerated through the same potential difference. Thereafter they enter normally in a region of uniform magnetic field and describes circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is:

[29 Jan, 2024 (Shift-II)]

- (a) $\left(\frac{R_2}{R_1}\right)^2$ (b) $\left(\frac{R_1}{R_2}\right)^2$ (c) $\left(\frac{R_1}{R_2}\right)$ (d) $\left(\frac{R_2}{R_1}\right)$

68. A charge of $4.0\mu\text{C}$ is moving with a velocity of $4.0 \times 10^6\text{ ms}^{-1}$ along the positive y-axis under a magnetic field $\vec{B}$ of strength $(2\hat{k})\text{ T}$. The force acting on the charge is $x\hat{i}\text{ N}$. The value of x is _____.

[29 Jan, 2024 (Shift-II)]

69. An electron moves through a uniform magnetic field $\vec{B} = B_0\hat{i} + 2B_0\hat{j}\text{ T}$. At a particular instant of time, the velocity of electron is $\vec{u} = 3\hat{i} + 5\hat{j}\text{ m/s}$. If the magnetic force acting on electron is $\vec{F} = 5ek\text{ N}$, where e is the charge of electron, then the value of B_0 is _____ T.

[31 Jan, 2024 (Shift-I)]

70. An electron is projected with uniform velocity along the axis inside a current carrying long solenoid. Then: [04 April, 2024 (Shift-I)]

- (a) the electron will be accelerated along the axis.
 (b) the electron will continue to move with uniform velocity along the axis of the solenoid.
 (c) the electron path will be circular about the axis.
 (d) the electron will experience a force at 45° to the axis and execute a helical path.

71. The electrostatic force ($\vec{F}_1$) and magnetic force ($\vec{F}_2$) acting on a charge q moving with velocity v can be written:

[05 April, 2024 (Shift-II)]

- (a) $\vec{F}_1 = q\vec{V} \cdot \vec{E}$, $\vec{F}_2 = q(\vec{B} \cdot \vec{V})$
 (b) $\vec{F}_1 = q\vec{B}$, $\vec{F}_2 = q(\vec{B} \times \vec{V})$
 (c) $\vec{F}_1 = q\vec{E}$, $\vec{F}_2 = q(\vec{V} \times \vec{B})$
 (d) $\vec{F}_1 = q\vec{E}$, $\vec{F}_2 = q(\vec{B} \times \vec{V})$
72. An electron with kinetic energy 5 eV enters a region of uniform magnetic field of 3 μ T perpendicular to its direction. An electric field E is applied perpendicular to the direction of velocity and magnetic field. The value of E , so that electron moves along the same path, is _____ NC^{-1} .
 (Given, mass of electron = 9×10^{-31} kg, electric charge = 1.6×10^{-19} C) [08 April, 2024 (Shift-I)]
73. A proton and a deuteron ($q = +e$, $m = 2.0u$) having same kinetic energies enter a region of uniform magnetic field $\vec{B}$, moving perpendicular to $\vec{B}$. The ratio of the radius r_d of deuteron path to the radius r_p of the proton path is: [09 April, 2024 (Shift-II)]
 (a) 1 : 1 (b) $1 : \sqrt{2}$ (c) $\sqrt{2} : 1$ (d) 1 : 2
74. A charge particle moving in magnetic field B , has the components of velocity along B as well as perpendicular to B . The path of the charge particle will be [8 April, 2023 (Shift-I)]
 (a) helical path with the axis perpendicular to the direction of magnetic field B
 (b) straight along the direction of magnetic field B
 (c) helical path with the axis along magnetic field B
 (d) circular path
75. An electron is moving along the positive x-axis. If the uniform magnetic field is applied parallel to the negative z-axis. then
 A. The electron will experience magnetic force along positive y-axis
 B. The electron will experience magnetic force along negative y-axis
 C. The electron will not experience any force in magnetic field
 D. The electron will continue to move along the positive x-axis
 E. The electron will move along circular path in magnetic field
 Choose the correct answer from the options given below:
 [13 April, 2023 (Shift-II)]
 (a) B and E only (b) A and E only
 (c) C and D only (d) B and D only
76. An electron is allowed to move with constant velocity along the axis of current carrying straight solenoid.
 A. The electron will experience magnetic force along the axis of the solenoid.
 B. The electron will not experience magnetic force.
 C. The electron will continue to move along the axis of the solenoid.
 D. The electron will be accelerated along the axis of the solenoid.
 E. The electron will follow parabolic path-inside the solenoid.
 Choose the correct answer from the options given below:
 [11 April, 2023 (Shift-II)]
 (a) B, C and D only (b) B and C only
 (c) A and D only (d) B and E only
77. A charge particle of 2 μ C accelerated by a potential difference of 100 V enters a region of uniform magnetic field of magnitude 4 mT at right angle to the direction of field. The charge particle completes semicircle of radius 3 cm inside magnetic field. The mass of the charge particle is _____ $\times 10^{-18}$ kg. [1 Feb, 2023 (Shift-I)]
78. A proton with a kinetic energy of 2.0 eV moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2} \times 10^{-3}$ T. The angle between

the direction of magnetic field and velocity of proton is 60° . The pitch of the helical path taken by the proton is _____ cm.
 (Take, mass of proton = 1.6×10^{-27} kg and Charge on proton = 1.6×10^{-19} C). [6 April, 2023 (Shift-II)]

79. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In an uniform magnetic field, speed and energy remains the same for a moving charged particle.
Reason (R): Moving charged particle experiences magnetic force perpendicular to its direction of motion. [24 June, 2022 (Shift-I)]
 (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (c) (A) is true but (R) is false.
 (d) (A) is false but (R) is true.
80. A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j})$ T. If it has an acceleration of $(\alpha\hat{i} - 4\hat{j}) \text{ m/s}^2$, then the value of α is [26 July, 2022 (Shift-I)]
 (a) 3 (b) 6 (c) 12 (d) 2
81. A proton, a deuteron and an α - particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is:
 [24 June, 2022 (Shift-II)]
 (a) $1 : \sqrt{2} : \sqrt{2}$ (b) $1 : 1 : \sqrt{2}$ (c) $\sqrt{2} : 1 : 1$ (d) $1 : \sqrt{2} : 1$
82. Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4, then, the ratio of their charges will be [25 July, 2022 (Shift-I)]
 (a) 8 : 5 (b) 5 : 4 (c) 5 : 3 (d) 8 : 7
83. An electron with energy 0.1 keV moves at right angle to the earth's magnetic field of $1 \times 10^{-4} \text{ Wbm}^{-2}$. The frequency of revolution of the electron will be
 (Take mass of electron = 9.0×10^{-31} kg) [25 July, 2022 (Shift-II)]
 (a) $66.6 \times 10^5 \text{ Hz}$ (b) $5.6 \times 10^5 \text{ Hz}$
 (c) $67.8 \times 10^6 \text{ Hz}$ (d) $1.8 \times 10^6 \text{ Hz}$
84. A proton and an alpha particle of the same velocity enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the radii of the circular paths described by the alpha particle and proton is: [26 June, 2022 (Shift-I)]
 (a) 1 : 4 (b) 4 : 1 (c) 2 : 1 (d) 1 : 2
85. Given below are two statements:
Statement-I: The electric force changes the speed of the charged particle and hence changes its kinetic energy; whereas the magnetic force does not change the kinetic energy of the charged particle.
Statement-II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field.
 In the light of the above statements, choose the most appropriate answer from the options given below: [29 June, 2022 (Shift-II)]
 (a) Both Statement-I and Statement-II are correct.
 (b) Both Statement-I and Statement-II are incorrect.
 (c) Statement-I is correct but Statement-II is incorrect.
 (d) Statement-I is incorrect but Statement-II is correct.
86. A beam of light travelling along X-axis is described by the electric field $E_y = 900 \sin \omega(t - x/c)$. The ratio of electric force to magnetic

force on a charge q moving along Y -axis with a speed of $3 \times 10^7 \text{ ms}^{-1}$ will be:

(Given speed of light = $3 \times 10^8 \text{ ms}^{-1}$) [27 July, 2022 (Shift-I)]

- (a) 1 : 1 (b) 1 : 10 (c) 10 : 1 (d) 1 : 2

97. A velocity selector consists of electric field $\vec{E} = E\hat{k}$ and magnetic field $\vec{B} = B\hat{j}$ with $B = 12 \text{ mT}$. The value of E required for an electron of energy 728 eV moving along the positive x -axis to pass undeflected is:

(Given. Mass of electron = $9.1 \times 10^{-31} \text{ kg}$) [26 July, 2022 (Shift-II)]

- (a) 192 kVm^{-1} (b) 192 mVm^{-1}
(c) 9600 kVm^{-1} (d) 16 kVm^{-1}

98. A 1 m long copper wire carries a current of 1 A. If the cross section of the wire is 2.0 mm^2 and the resistivity of copper is $1.7 \times 10^{-8} \Omega \text{ m}$, the force experienced by moving electron in the wire is _____ $\times 10^{-23} \text{ N}$.

(charge on electron = $-1.6 \times 10^{-19} \text{ C}$) [27 July, 2022 (Shift-I)]

99. A deuteron and a proton moving with equal kinetic energy enter into a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{r_d}{r_p}$ will be $\sqrt{x} : 1$ where x is.....

[27 June, 2022 (Shift-II)]

100. A singly ionized magnesium atom ($A = 24$) ion is accelerated to kinetic energy 5 keV, and is projected perpendicularly into a magnetic field B of the magnitude 0.5 T. The radius of path formed will be _____ cm.

[28 June, 2022 (Shift-I)]

101. A deuteron and an alpha particle having equal kinetic energy enter perpendicularly into a magnetic field. Let r_d and r_α be their respective radii of circular path. The value of $\frac{r_d}{r_\alpha}$ is equal to:

[20 July, 2021 (Shift-I)]

- (a) $\sqrt{2}$ (b) 2 (c) $\frac{1}{\sqrt{2}}$ (d) 1

102. A proton and an α -particle, having kinetic energy K_p and K_α respectively, enter into a magnetic field at right angles. The ratio of the radii of trajectory of proton to that of α -particle is 2 : 1. The ratio of $K_p : K_\alpha$ is

[18 March, 2021 (Shift-II)]

- (a) 1 : 8 (b) 8 : 1 (c) 1 : 4 (d) 4 : 1

103. Two ions of masses 4 amu and 16 amu have charges $+2e$ and $+3e$ respectively. These ions pass through the region of constant perpendicular magnetic field. The kinetic energy of both ions is same. Then:

[27 Aug, 2021 (Shift-I)]

- (a) No ion will be deflected.
(b) Both ions will be deflected equally
(c) Lighter ion will be deflected more than heavier ion
(d) Lighter ion will be deflected less than heavier ion

104. A charge Q is moving $d\vec{l}$ distance in the magnetic field $\vec{B}$. Find the value of work done by $\vec{B}$.

[16 March, 2021 (Shift-II)]

- (a) 1 (b) Zero (c) Infinite (d) -1

105. Two ions having same mass have charges in the ratio 1 : 2. They are projected normally in a uniform magnetic field with their speeds in the ratio 2 : 3. The ratio of the radii of their circular trajectories is:

[25 July, 2021 (Shift-II)]

- (a) 3 : 1 (b) 2 : 3 (c) 4 : 3 (d) 1 : 4

106. A proton, a deuteron and an α -particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces acting on them is _____ and their speed is _____ in the ratio.

[25 Feb, 2021 (Shift-I)]

- (a) 4 : 2 : 1 and 2 : 1 : 1 (b) 2 : 1 : 1 and 4 : 2 : 1
(c) 1 : 2 : 4 and 1 : 1 : 2 (d) 1 : 2 : 4 and 2 : 1 : 1

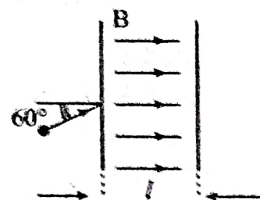
97. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0\hat{j}$. If an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{i}$ act on the particle, its speed will double after a time [7 Jan, 2020 (Shift-II)]

- (a) $\frac{2mv_0}{qE_0}$ (b) $\frac{3mv_0}{qE_0}$ (c) $\frac{\sqrt{2}mv_0}{qE_0}$ (d) $\frac{\sqrt{3}mv_0}{qE_0}$

98. An electron is moving along $+x$ direction with a velocity of $6 \times 10^6 \text{ ms}^{-1}$. It enters a region of uniform electric field of 300 V/cm pointing along $+y$ direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be: [6 Sep, 2020 (Shift-I)]

- (a) $3 \times 10^{-4} \text{ T}$, along $+z$ direction
(b) $5 \times 10^{-3} \text{ T}$, along $-z$ direction
(c) $5 \times 10^{-3} \text{ T}$, along $+z$ direction
(d) $3 \times 10^{-4} \text{ T}$, along $-z$ direction

99. The figure shows a region of length ' l ' with a uniform magnetic field of 0.3 T in it and a proton entering the region with velocity $4 \times 10^5 \text{ ms}^{-1}$ making an angle 60° with the field. If the proton completes 10 revolution by the time it cross the region shown, ' l ' is close to (mass of proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.6 \times 10^{-19} \text{ C}$) [2 Sep, 2020 (Shift-II)]



- (a) 0.44 m (b) 0.11 m (c) 0.88 m (d) 0.22 m

100. A charged particle carrying charge $1 \mu\text{C}$ is moving with velocity $(2\hat{i} + 3\hat{j} + 4\hat{k})\text{ms}^{-1}$. If an external magnetic field of $(5\hat{i} + 3\hat{j} - 6\hat{k}) \times 10^{-4} \text{ T}$ exists in the region where the particle is moving then the force on the particle is $\vec{F} \times 10^{-4} \text{ N}$. The vector $\vec{F}$ is

[3 Sep, 2020 (Shift-I)]

- (a) $-3.0\hat{i} + 3.2\hat{j} - 0.9\hat{k}$ (b) $-300\hat{i} + 320\hat{j} - 90\hat{k}$
(c) $-0.30\hat{i} + 0.32\hat{j} - 0.09\hat{k}$ (d) $-30\hat{i} + 32\hat{j} - 9\hat{k}$

101. A particle of charge q and mass m is moving with a velocity $-vi$ ($v \neq 0$) towards a large screen placed in the $Y-Z$ plane at a distance d . If there is a magnetic field $\vec{B} = B_0\hat{x}$, the minimum value of v for which the particle will not hit the screen is:

[6 Sep, 2020 (Shift-I)]

- (a) $\frac{qdB_0}{3m}$ (b) $\frac{2qdB_0}{m}$ (c) $\frac{qdB_0}{m}$ (d) $\frac{qdB_0}{2m}$

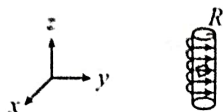
102. Proton with kinetic energy of 1 MeV moves from south to north. It gets an acceleration of 10^{12} m/s^2 by an applied magnetic field (west to east). The value of magnetic field. (Rest mass of proton is $1.6 \times 10^{-27} \text{ kg}$) [8 Jan, 2020 (Shift-I)]

- (a) 7.1 mT (b) 71 mT (c) 0.71 mT (d) 0.071 mT

103. A beam of protons with speed $4 \times 10^5 \text{ m/s}$ enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. The pitch of the resulting helical path of protons is close to (Mass of the proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.6 \times 10^{-19} \text{ C}$) [2 Sep, 2020 (Shift-I)]

- (a) 2 cm (b) 12 cm (c) 5 cm (d) 4 cm

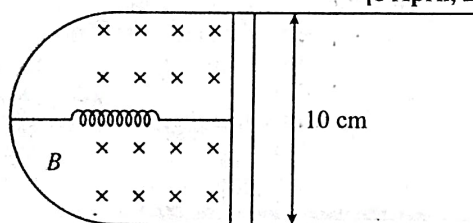
104. An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I . The electron gun shoots an electron along the radius of the solenoid with speed v . If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning) [9 Jan, 2020 (Shift-II)]



- (a) $\frac{e\mu_0 nIR}{2m}$ (b) $\frac{e\mu_0 nIR}{m}$ (c) $\frac{2e\mu_0 nIR}{m}$ (d) $\frac{e\mu_0 nIR}{4m}$

105. A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constants 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T . If the strip is pulled from its equilibrium position and released, the number of oscillation it performs before its amplitude decreases by a factor of e is N . If the mass of the strip is 50 grams, its resistance 10Ω and air drag negligible, N will be close to

[8 April, 2019 (Shift-I)]



- (a) 50000 (b) 5000 (c) 10000 (d) 1000

106. A moving coil galvanometer has a coil with 175 turns and area 1 cm^2 . It uses a torsion band of torsion constant 10^{-6} N-m/rad . The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1 mA . The value of B (in Tesla) is approximately [9 April, 2019 (Shift-II)]

- (a) 10^{-3} (b) 10^{-1} (c) 10^{-4} (d) 10^{-2}

107. A proton, an electron, and a Helium nucleus, have the same energy. They are in circular orbitals in a plane due to magnetic field perpendicular to the plane. Let r_p , r_e and r_{He} be their respective radii, then, [10 April, 2019 (Shift-I)]

- (a) $r_e > r_p > r_{He}$ (b) $r_e < r_p < r_{He}$
(c) $r_e < r_p = r_{He}$ (d) $r_e > r_p = r_{He}$

108. The region between $y=0$ and $y=d$ contains a magnetic field $\vec{B} = B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v} = v\hat{i}$. If $d = \frac{mv}{2qB}$, the acceleration of the charged particle at the point of its emergence at the other side is: [11 Jan, 2019 (Shift-II)]

- (a) $\frac{qvB}{m} \left(\frac{1}{2}\hat{j} - \frac{\sqrt{3}}{2}\hat{j} \right)$ (b) $\frac{qvB}{m} \left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j} \right)$
(c) $\frac{qvB}{m} \left(\frac{-\hat{j} + \hat{i}}{\sqrt{2}} \right)$ (d) $\frac{qvB}{m} \left(\frac{\hat{j} + \hat{i}}{\sqrt{2}} \right)$

109. A proton and an α -particle (with their masses in the ratio of 1 : 4 and charges in the ratio of 1 : 2) are accelerated from rest through a potential difference V . If a uniform magnetic field (B) is set up perpendicular to their velocities, the ratio of the radii r_p : r_α of the circular paths described by them will be [12 Jan, 2019 (Shift-I)]

- (a) $1:\sqrt{2}$ (b) $1:2$ (c) $1:3$ (d) $1:\sqrt{3}$

110. In an experiment, electrons are accelerated, from rest, by applying a voltage of 500 V . Calculate the radius of the path if a magnetic field 100 mT is then applied. [Charge of the electron = $1.6 \times 10^{-19} \text{ C}$ Mass of the electron = $9.1 \times 10^{-31} \text{ kg}$] [11 Jan, 2019 (Shift-I)]

- (a) $7.5 \times 10^{-3} \text{ m}$ (b) $7.5 \times 10^{-2} \text{ m}$
(c) 7.5 m (d) $7.5 \times 10^{-4} \text{ m}$

Magnetic Force on Current Carrying Wire

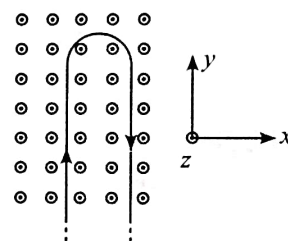
111. A 4.0 cm long straight wire carrying a current of 8 A is placed perpendicular to an uniform magnetic field of strength 0.15 T . The magnetic force on the wire is _____ mN .

[03 April, 2025 (Shift-I)]

112. The horizontal component of earth's magnetic field at a place is $3.5 \times 10^{-5} \text{ T}$. A very long straight conductor carrying current of $\sqrt{2} \text{ A}$ in the direction from South east to North West is placed. The force per unit length experienced by the conductor is _____ $\times 10^{-6} \text{ N/m}$.

[30 Jan, 2024 (Shift-I)]

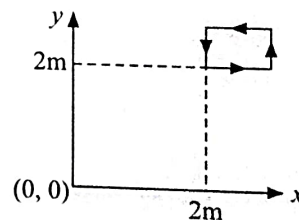
113. A rigid wire consists of a semicircular portion of radius R and two straight sections. The wire is partially immersed in a perpendicular magnetic field $B = B_0 \hat{j}$ as shown in figure. The magnetic force on the wire if it has a current i is: [31 Jan, 2024 (Shift-I)]



- (a) $-iBR \hat{j}$ (b) $2iBR \hat{j}$ (c) $iBR \hat{j}$ (d) $-2iBR \hat{j}$

114. The magnetic field existing in a region is given by $\vec{B} = 0.2(1 + 2x)\hat{k} \text{ T}$.

A square loop of edge 50 cm carrying 0.5 A current is placed in x - y plane with its edges parallel to the x - y axes, as shown in figure. The magnitude of the net magnetic force experienced by the loop is _____ mN .



[04 April, 2024 (Shift-I)]

115. A 2 A current carrying straight metal wire of resistance 1Ω , resistivity $2 \times 10^{-6} \Omega \text{ m}$, area of cross-section 10 mm^2 and mass 500 g is suspended horizontally in mid air by applying a uniform magnetic field B . The magnitude of B is _____ $\times 10^{-1} \text{ T}$ (given, $g = 10 \text{ m/s}^2$) [05 April, 2024 (Shift-I)]

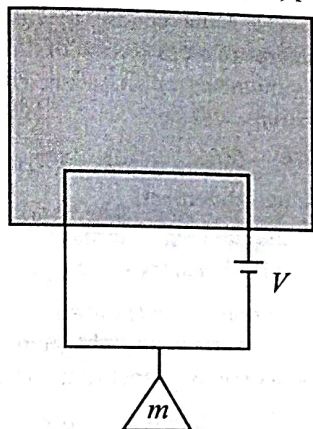
116. A square loop of edge length 2 m carrying current of 2 A is placed with its edges parallel to the x - y axis. A magnetic field is passing through the x - y plane and expressed as $\vec{B} = B_0(1 + 4x)\hat{k}$, where $B_0 = 5 \text{ T}$. The net magnetic force experienced by the loop is _____ N .

[09 April, 2024 (Shift-I)]

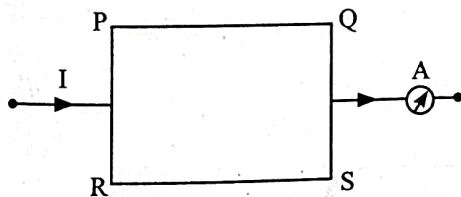
117. Two long straight wires P and Q carrying equal current of $10A$ each were kept parallel to each other at a distance of $5cm$. Magnitude of magnetic force experienced by $10cm$ length of wire P is F_1 . If distance between the wires is halved and currents on them are doubled, force F_2 on $10 cm$ length of wire P will be:

[24 Jan, 2023 (Shift-I)]

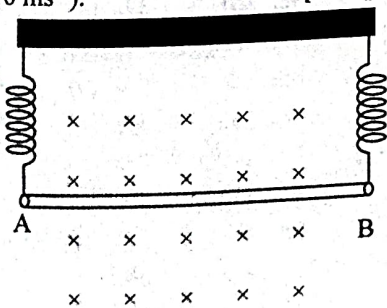
- (a) $8 F_1$ (b) $10 F_1$ (c) $F_1/8$ (d) $F_1/10$
118. A massless square loop, of wire of resistance 10Ω , supporting a mass of $1 g$, hangs vertically with one of its sides in a uniform magnetic field of $10^3 G$, directed outwards in the shaded region. A dc voltage V is applied to the loop. For what value of V , the magnetic force will exactly balance the weight of the supporting mass of $1 g$? (If sides of the loop = $10cm$, $g = 10 ms^{-2}$) [30 Jan, 2023 (Shift-I)]



- (a) $\frac{1}{10} V$ (b) $100 V$ (c) $1 V$ (d) $10 V$
119. A current carrying rectangular loop PQRS is made of uniform wire. The length $PR = QS = 5cm$ and $QR = PS = 100 cm$. If ammeter current reading changes from I to $2I$, the ratio of magnetic forces per unit length on the wire PQ due to wire RS in the two cases respectively $f_{PQ}^1 : f_{PQ}^{2I}$ is: [30 Jan, 2023 (Shift-II)]



- (a) 1 : 2 (b) 1 : 4 (c) 1 : 5 (d) 1 : 3
120. A straight wire AB of mass $40 g$ and length $50 cm$ is suspended by a pair of flexible leads in uniform magnetic field of magnitude $0.40 T$ as shown in the figure. The magnitude of the current required in the wire to remove the tension in the supporting leads is _____ A. (Take $g = 10 ms^{-2}$). [13 April, 2023 (Shift-II)]



121. A particle of mass m and charge q is in an electric and magnetic field given by:
 $\vec{E} = 2\hat{i} + 3\hat{j}$; $\vec{B} = 4\hat{j} + 6\hat{k}$

The charged particle is shifted from the origin to the point $P(x = 1; y = 1)$ along a straight path. The magnitude of the total work done is:

[11 Jan, 2019 (Shift-II)]

- (a) $(0.35)q$ (b) $5q$ (c) $(2.5)q$ (d) $(0.15)q$

Torque and Potential Energy of Coil in External Magnetic Field, Current Sensitivity

122. A solenoid having area A and length ' l ' is filled with a material having relative permeability 2. The magnetic energy stored in the solenoid is: [02 April, 2025 (Shift-II)]

- (a) $\frac{B^2 A l}{\mu_0}$ (b) $\frac{B^2 A l}{2\mu_0}$
 (c) $B^2 A l$ (d) $\frac{B^2 A l}{4\mu_0}$

123. In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars:

$$R_1 = 5 \Omega, N_1 = 15, A_1 = 3.6 \times 10^{-3} m^2, B_1 = 0.25 T$$

$$R_2 = 7 \Omega, N_2 = 21, A_2 = 1.8 \times 10^{-3} m^2, B_2 = 0.50 T$$

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of voltage sensitivity of M_1 and M_2 ?

[02 April, 2025 (Shift-II)]

- (a) 1 : 1 (b) 1 : 4
 (c) 1 : 3 (d) 1 : 2

124. Consider a moving coil galvanometer (MCG):

[23 Jan, 2025 (Shift-I)]

- The torsional constant in moving coil galvanometer has dimensions $[ML^2 T^{-2}]$
- Increasing the current sensitivity may not necessarily increase the voltage sensitivity.
- If we increase number of turns (N) to its double ($2N$), then the voltage sensitivity doubles.
- MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.
- Current sensitivity of MCG depends inversely on number of turns of coil.

Choose the correct answer from the options given below:

- (a) B, D, E Only (b) A, D Only
 (c) A, B, E Only (d) A, B Only

125. A current of $200 \mu A$ deflects the coil of a moving coil galvanometer through 60° . The current to cause deflection through $\frac{\pi}{10}$ radian is:

[27 Jan, 2024 (Shift-II)]

- (a) $30 \mu A$ (b) $120 \mu A$ (c) $60 \mu A$ (d) $180 \mu A$

126. The deflection in moving coil galvanometer falls from 25 divisions to 5 division when a shunt of 24Ω is applied. The resistance of galvanometer coil will be:

[29 Jan, 2024 (Shift-I)]

- (a) 12Ω (b) 96Ω (c) 48Ω (d) 100Ω

127. A uniform magnetic field of $2 \times 10^{-3} T$ acts along positive Y-direction. A rectangular loop of sides $20 cm$ and $10 cm$ with current of $5 A$ is in Y-Z plane. The current is in anticlockwise sense with reference to negative X axis. Magnitude and direction of the torque is:

[31 Jan, 2024 (Shift-II)]

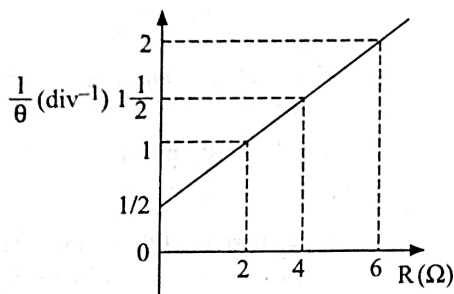
- (a) 2×10^{-4} N-m along positive Z-direction
 (b) 2×10^{-4} N-m along negative Z-direction
 (c) 2×10^{-4} N-m along positive X-direction
 (d) 2×10^{-4} N-m along positive Y-direction

128. A moving coil galvanometer has 100 turns and each turn has an area of 2.0 cm^2 . The magnetic field produced by the magnet is 0.01 T and the deflection in the coil is 0.05 radian when a current of 10 mA is passed through it. The torsional constant of the suspension wire is $x \times 10^{-5} \text{ N-m/rad}$. The value of x is _____.

[1 Feb, 2024 (Shift-II)]

129. In the experiment to determine the galvanometer resistance by half-deflection method, the plot of $1/\theta$ vs the resistance (R) of the resistance box is shown in the figure. The figure of merit of the galvanometer is $\times 10^{-1} \text{ A/division}$. [The source has emf 2 V]

[05 April, 2024 (Shift-I)]



130. A circular coil having 200 turns, $2.5 \times 10^{-4} \text{ m}^2$ area and carrying $100 \mu\text{A}$ current is placed in a uniform magnetic field of 1 T . Initially the magnetic dipole moment ($\vec{M}$) was directed along $\vec{B}$. Amount of work, required to rotate the coil through 90° from its initial orientation such that $\vec{M}$ becomes perpendicular to B , is _____ μJ .

[06 April, 2024 (Shift-I)]

131. A coil having 100 turns, area of $5 \times 10^{-3} \text{ m}^2$, carrying current of 1 mA is placed in uniform magnetic field of 0.20 T such a way that plane of coil is perpendicular to the magnetic field. The work done in turning the coil through 90° is _____ μJ . [06 April, 2024 (Shift-II)]

132. Certain galvanometers have a fixed core made of non magnetic metallic material. The function of this metallic material is _____ [8 April, 2023 (Shift-I)]

- (a) to oscillate the coil in magnetic field for longer period of time
 (b) to bring the coil to rest quickly
 (c) to produce large deflecting torque on the coil
 (d) to make the magnetic field radial

133. The current sensitivity of moving coil galvanometer is increased by 25% . This increase is achieved only by changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be: [11 April, 2023 (Shift-I)]

- (a) $+25\%$ (b) -50% (c) Zero (d) -25%

134. Given below are two statement:

Statement-I : If the number of turns in the coil of a moving coil galvanometer is doubled then the current sensitivity becomes double.

Statement-II : Increasing current sensitivity of a moving coil galvanometer by only increasing the number of turns in the coil will also increase its voltage sensitivity in the same ratio:

In the light of the above statement, choose the correct answer from the options given below: [10 April, 2023 (Shift-I)]

- (a) Statement-I is false but Statement-II is true
 (b) Both Statement-I and Statement-II are true
 (c) Both Statement-I and Statement-II are false
 (d) Statement-I is true but Statement-II is false

ANSWER KEY

- | | | | | | | | | | |
|------------|-----------|----------|-----------|-----------|------------|-----------|-----------|----------|-----------|
| 1. (a) | 2. [1] | 3. [1] | 4. [8] | 5. (b) | 6. [160] | 7. [3.00] | 8. (c) | 9. [40] | 10. [72] |
| 11. [5] | 12. [2] | 13. [2] | 14. (d) | 15. (c) | 16. (d) | 17. (c) | 18. (d) | 19. (c) | 20. (c) |
| 21. [68] | 22. [628] | 23. [2] | 24. [40] | 25. [8] | 26. (b) | 27. (c) | 28. (c) | 29. (b) | 30. (b) |
| 31. (d) | 32. (c) | 33. (d) | 34. (b) | 35. (d) | 36. (c) | 37. (d) | 38. (d) | 39. (a) | 40. (d) |
| 41. (c) | 42. (c) | 43. (c) | 44. (a) | 45. [250] | 46. (d) | 47. [8] | 48. [20] | 49. (c) | 50. [500] |
| 51. (c) | 52. (b) | 53. (b) | 54. (c) | 55. (a) | 56. (d) | 57. (d) | 58. (d) | 59. (a) | 60. [0] |
| 61. (d) | 62. (d) | 63. [2] | 64. (b) | 65. (d) | 66. (c) | 67. (b) | 68. [32] | 69. [5] | 70. (b) |
| 71. (c) | 72. (d) | 73. (c) | 74. (c) | 75. (a) | 76. (b) | 77. [144] | 78. [40] | 79. (a) | 80. (b) |
| 81. (d) | 82. (b) | 83. (c) | 84. (c) | 85. (c) | 86. (c) | 87. (a) | 88. [136] | 89. [2] | 90. [10] |
| 91. (a) | 92. (d) | 93. (c) | 94. (b) | 95. (c) | 96. (a) | 97. (d) | 98. (c) | 99. (a) | 100. (d) |
| 101. (c) | 102. (c) | 103. (d) | 104. (a) | 105. (b) | 106. (a) | 107. (c) | 108. (*) | 109. (a) | 110. (d) |
| 111. [48] | 112. [35] | 113. (d) | 114. [50] | 115. [5] | 116. [160] | 117. (a) | 118. (d) | 119. (b) | 120. [2] |
| 121. (b) | 122. (d) | 123. (a) | 124. (d) | 125. (c) | 126. (b) | 127. (b) | 128. [4] | 129. [5] | 130. [2] |
| 131. [100] | 132. (b) | 133. (a) | 134. (d) | | | | | | |